

## CO3133 — Deep Learning and Its Applications

### Assignment 2 Dataset Proposal

#### 1. Proposed Task

**Task:** Multi-class image classification

**Application:** Rice variety identification from individual grain images.

The objective is to develop a deep-learning model capable of classifying an input image of a rice grain into one of five rice varieties:

- Arborio
- Basmati
- Ipsala
- Jasmine
- Karacadag

The project will investigate whether a targeted architectural modification to a convolutional neural network can improve classification performance while maintaining reasonable computational cost.

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#### 2. Dataset

**Dataset:** Rice Image Dataset

**Source:** Kaggle — Murat Koklu Dataset

**Year:** 2021

**Dataset URL:** <https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset>

**Original dataset:** Koklu et al., *Classification of rice varieties with deep learning methods*.

**License:** The dataset is publicly available; the project will retain the original dataset attribution and verify the applicable license terms before redistribution ([CC0: Public Domain](#)).

The dataset contains **75,000 RGB images** of individual rice grains, with **15,000 images per class**. The five classes are Arborio, Basmati, Ipsala, Jasmine, and Karacadag. The images are approximately **250 × 250 pixels** and are provided in JPEG format.

| Class        | Number of images |
|--------------|------------------|
| Arborio      | 15,000           |
| Basmati      | 15,000           |
| Ipsala       | 15,000           |
| Jasmine      | 15,000           |
| Karacadag    | 15,000           |
| <b>Total</b> | <b>75,000</b>    |

The dataset is therefore balanced across all five classes.

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### 3. Task Input, Output and Annotation Type

**Input:** An RGB image containing an individual rice grain.

**Output:** A probability distribution over five rice-variety classes.

**Annotation Type:** Single-label image classification. Each image is annotated with one rice variety label: Arborio, Basmati, Ipsala, Jasmine, or Karacadag.

**The predicted class will be the rice variety with the highest predicted probability.**

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### 4. Preliminary Data Analysis

The dataset is balanced, with exactly 15,000 images available for each class. This removes the need for class-weighted loss solely to compensate for class imbalance.

The images contain individual rice grains captured under a controlled imaging setup. The relevant visual characteristics for distinguishing varieties include differences in grain shape, size, color, and texture.



*Figure 1. A sample of Arborio rice*



*Figure 2. A sample of Basmati rice*

Before model training, the following exploratory analysis will be performed:

1. Verify the number of samples in every class.
2. Check for corrupted or unreadable images.
3. Analyze image dimensions and color channels.
4. Visualize representative samples from every class.
5. Examine intra-class and inter-class visual variation.
6. Check for duplicate or near-duplicate images where practical.

7. Analyze whether the dataset contains visually similar samples that could lead to train/test leakage.

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## 5. Data Split and Leakage Prevention

The dataset will be divided into:

- **70% training**
- **15% validation**
- **15% test**

The split will be performed **stratified by rice variety**, preserving approximately equal class proportions in all three subsets.

A fixed random seed will be used to make the split reproducible.

Because the dataset consists of individual grain images and does not provide an explicit subject/identity grouping variable, the primary split unit will be the **individual image**. Before finalizing the split, duplicate and near-duplicate images will be investigated to reduce the possibility of information leakage between training, validation, and test sets.

The test set will remain untouched during model development and will only be used for final evaluation.

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## 6. Evaluation Metrics

The primary evaluation metrics will be:

- **Accuracy**
- **Macro-F1 score**
- **FLOPs (optional)**

Accuracy is appropriate because the five classes are balanced. Macro-F1 will additionally measure performance independently across the five classes and will prevent strong performance on some classes from completely masking poor performance on another class.

FLOPs measures the computational cost of the models.

A confusion matrix and per-class precision, recall, and F1-score will also be reported for detailed error analysis.

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## 7. Baseline Model

A simple convolutional neural network will be implemented as a baseline.

The baseline will consist of several convolutional blocks followed by global pooling and a classification head. It will be trained directly on the rice image dataset.

The purpose of this baseline is to establish the performance obtainable without using a large pretrained backbone and to provide a reference point for evaluating the more advanced models.

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## 8. Pretrained / Modern Model

A pretrained **MobileNetV3** model will be used as the primary modern model.

The model will use weights pretrained on ImageNet, with its final classification layer replaced by a five-class classification head.

The model will initially be fine-tuned on the Rice Image Dataset. The fine-tuning protocol will be documented, including:

- pretrained initialization;
- input preprocessing;
- replacement of the classification head;
- learning rate;
- optimizer;
- batch size;
- number of epochs;
- checkpoint selection criterion;
- random seed.

A comparison between partial freezing and full fine-tuning may also be considered depending on computational cost.

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## 9. Proposed Architectural Experiment

The main research question of the project is:

## **Can a targeted modification to the CNN architecture improve rice-variety classification compared with the original architecture while maintaining reasonable computational cost?**

The pretrained MobileNetV3 will serve as the main reference architecture. A selected architectural component will then be modified while keeping the dataset, preprocessing, training protocol, and evaluation procedure as consistent as possible.

The controlled experiment will compare:

### **Model A — Reference:**

Original MobileNetV3 architecture with a five-class classification head.

### **Model B — Modified:**

MobileNetV3 with the selected architectural modification.

The exact modification will be finalized after preliminary experimentation and literature review. Candidate modifications may include replacing a convolutional block, changing the feature-extraction layer, modifying the pooling mechanism, or introducing an alternative feature-processing component.

The hypothesis is that the architectural modification will produce a measurable improvement in classification performance by allowing the network to extract more discriminative features from rice-grain shape, texture, and appearance.

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## **10. Controlled Experiment**

The primary controlled experiment will follow the principle of changing **one architectural factor at a time**.

The following factors will remain fixed between the reference and modified models:

- Dataset
- Train/validation/test split
- Image preprocessing
- Data augmentation
- Optimizer
- Training epochs
- Batch size

- Learning-rate schedule
- Random seed
- Model-selection criterion
- Evaluation metrics

The primary changed factor will be the selected architectural component.

The models will be compared using:

- Accuracy
- Macro-F1
- Per-class F1
- Number of trainable parameters
- Training time
- Inference cost
- FLOPs

This will allow the project to determine whether an improvement in predictive performance justifies any additional computational cost.

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## **11. Qualitative Evaluation and Error Analysis**

In addition to numerical evaluation, the project will analyze incorrectly classified rice grains.

Examples will be grouped according to error patterns, such as:

- visually similar rice varieties;
- unusual grain shapes;
- differences in orientation;
- ambiguous texture or color;
- preprocessing or image-quality issues.

Confusion matrices will be used to identify which rice varieties are most frequently confused with one another.

Representative correct and incorrect predictions will be visualized to investigate the behavior of the models.

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## **12. Compute Estimate**

The experiments will be conducted using GPU acceleration.

The full dataset contains 75,000 images, so training will use mini-batches and standard image resizing appropriate for the selected pretrained architecture.

The proposal will initially allocate approximately:

- **1 GPU**
- **8 – 16GB GPU memory preferred**
- **Several hours per major training configuration**

The exact training time, hardware specification, software versions, parameter count, and inference time will be recorded during the experiments.

If computational resources become limiting, a subset may be used during preliminary experimentation, while the final comparison will use the full approved dataset where feasible.

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## **13. Expected Outcome**

The project is expected to establish whether the proposed CNN architectural modification provides a meaningful improvement over the reference architecture for rice-variety classification.

The final analysis will compare:

1. The simple CNN baseline.
2. The pretrained MobileNetV3 reference model.
3. The modified MobileNetV3 model.

The results will be evaluated not only by classification performance but also by computational cost and error characteristics.



The main goal is therefore not simply to maximize accuracy, but to determine whether the proposed architectural change provides a reproducible and meaningful improvement under controlled experimental conditions.